

Semigroup Spectral Extraction from Euclidean Exponential Clustering

Companion note: OS reconstruction and mass-gap criteria on a bounded macroscopic net

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Abstract

This note records a self-contained “spectral extraction” mechanism used in macroscopic mass-gap constructions: exponential decay of Euclidean time-separated correlations, together with Osterwalder–Schrader (OS) positivity on a bounded local net of observables, yields a strictly positive spectral gap for the reconstructed Hamiltonian. The emphasis is structural. The spectral-gap conclusion is derived from semigroup spectral theory and does not require any global renormalization-group flow analysis or stable-manifold machinery. The statements are formulated on a bounded macroscopic observable net at fixed scale $\ell > 0$ (no sharp $t \rightarrow 0$ local fields).

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1 Purpose and scope

The technical Route 1 construction establishes, at a fixed macroscopic scale $\ell > 0$, two Euclidean outputs on a bounded local net of gauge-invariant observables: (i) Osterwalder–Schrader positivity

(reflection positivity) and (ii) exponential clustering of time-separated correlations (typically obtained from Dobrushin-type contractivity; cf. [1, 2]). This companion note isolates the logically minimal bridge from these Euclidean properties to a Minkowski-space spectral statement: a strictly positive lower bound on the spectrum of the reconstructed Hamiltonian above the vacuum.

The key point is that the relevant “time” parameter is the OS semigroup time in the reconstruction, not an RG flow parameter. Once OS positivity is available, Euclidean time translations generate a positivity-preserving semigroup e^{-tH} on the reconstructed Hilbert space. Exponential Euclidean decay becomes an exponential bound on semigroup matrix elements. A spectral-theorem argument then excludes spectrum below the corresponding decay rate.

Remark 1.1 (What is and is not addressed). All statements are made on a bounded macroscopic observable net $\mathcal{O}_\ell$ at fixed scale $\ell > 0$. This note does *not* construct sharp local operator-valued distributions obtained by taking a $t \rightarrow 0$ limit, and it does not address the ultraviolet completion problem beyond the hypotheses stated explicitly.

Remark 1.2 (Terminology: “mass gap” in this note). Throughout, “mass gap” refers to a *spectral gap* for the reconstructed Hamiltonian H above the vacuum:

$$\text{Spec}(H) \subset \{0\} \cup [m, \infty).$$

This is the notion naturally extracted from OS positivity and Euclidean time clustering on the bounded macroscopic net. Questions about sharper ultraviolet localization (e.g. $t \rightarrow 0$ fields) or finer energy–momentum spectral structure lie outside the present scope.

Remark 1.3 (Vacuum sector). The spectral conclusion in this note excludes spectrum in $(0, m)$ and therefore provides a *gap above the ground-state sector*. In the OS reconstruction, the distinguished vector $\Omega = [1]$ satisfies $H\Omega = 0$. Under the exponential clustering hypothesis used for spectral extraction (Definition 5.1), one in fact obtains *uniqueness of the ground state* in the reconstructed representation (see Corollary 5.6 below). No claim is made here about the full relativistic energy–momentum spectrum.

2 OS setup on a bounded macroscopic net

We work in Euclidean space $\mathbb{R}^4$ with coordinates $x = (t, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^3$. Let $\theta(t, \mathbf{x}) = (-t, \mathbf{x})$ denote time reflection.

2.1 Bounded local net

Definition 2.1 (Bounded macroscopic net). Fix a macroscopic scale $\ell > 0$. A *bounded macroscopic observable net* is an assignment

$$\Lambda \mapsto \mathcal{O}_\ell(\Lambda)$$

from bounded Borel sets $\Lambda \subset \mathbb{R}^4$ to unital $*$ -algebras of bounded observables, such that:

- (N1) **Isotony.** If $\Lambda_1 \subset \Lambda_2$ then $\mathcal{O}_\ell(\Lambda_1) \subset \mathcal{O}_\ell(\Lambda_2)$.
- (N2) **Translation covariance.** There is an action $(\tau_a)_{a \in \mathbb{R}^4}$ by $*$ -automorphisms such that $\tau_a \mathcal{O}_\ell(\Lambda) = \mathcal{O}_\ell(\Lambda + a)$.
- (N3) **Geometric time reflection.** There is an involutive $*$ -automorphism ι_θ such that

$$\iota_\theta \mathcal{O}_\ell(\Lambda) = \mathcal{O}_\ell(\theta\Lambda), \quad \iota_\theta^2 = \text{id}, \quad \iota_\theta \circ \tau_a = \tau_{\theta a} \circ \iota_\theta.$$

Define the global algebra $\mathcal{A} := \bigcup_{\Lambda} \mathcal{O}_{\ell}(\Lambda)$ and the *OS reflection* $\Theta : \mathcal{A} \rightarrow \mathcal{A}$ by

$$\Theta(F) := \iota_{\theta}(F^*).$$

Then Θ is an antilinear involution and an anti-automorphism: $\Theta(FG) = \Theta(G)\Theta(F)$, and $\Theta\mathcal{O}_{\ell}(\Lambda) = \mathcal{O}_{\ell}(\theta\Lambda)$.

Definition 2.2 (Positive-time algebra). Let $\mathbb{R}_+^4 := \{(t, \mathbf{x}) \in \mathbb{R}^4 : t > 0\}$. Define the *positive-time algebra*

$$\mathcal{A}_+ := \bigcup_{\Lambda \subset \mathbb{R}_+^4 \text{ bounded}} \mathcal{O}_{\ell}(\Lambda) \subset \mathcal{A}.$$

We write $\tau_t := \tau_{(t, \mathbf{0})}$ for Euclidean time translations.

2.2 OS-positive functional

Definition 2.3 (OS-positive Schwinger functional on $\mathcal{A}$). A linear functional $\mathcal{S} : \mathcal{A} \rightarrow \mathbb{C}$ is an *OS-positive, translation-invariant Schwinger functional* if:

(OS1) ***-compatibility.** $\mathcal{S}(F^*) = \overline{\mathcal{S}(F)}$ for all $F \in \mathcal{A}$.

(OS2) **Normalization.** $\mathcal{S}(1) = 1$.

(OS3) **Reflection positivity.** For all $F \in \mathcal{A}_+$,

$$\mathcal{S}(\Theta(F) F) \geq 0.$$

(OS4) **Translation invariance.** $\mathcal{S}(\tau_a F) = \mathcal{S}(F)$ for all $a \in \mathbb{R}^4$ and $F \in \mathcal{A}$.

(OS5) **Time-regularity (weak continuity).** For all $F, G \in \mathcal{A}_+$, the map

$$t \mapsto \mathcal{S}(\Theta(F) \tau_t G)$$

is continuous on $[0, \infty)$.

Remark 2.4 (Compatibility of net and OS data). The OS time-translation semigroup arises by restricting the full Euclidean translation action to the time axis: $\tau_t = \tau_{(t, \mathbf{0})}$. The positive-time algebra $\mathcal{A}_+$ is defined from the net by localization in $\mathbb{R}_+^4$. The OS reflection is $\Theta = \iota_{\theta} \circ (\cdot)^*$.

3 OS reconstruction and the semigroup

We record the standard bounded-net OS reconstruction steps (see, e.g., Osterwalder–Schrader and Glimm–Jaffe).

3.1 Hilbert space and vacuum

Definition 3.1 (OS inner product and Hilbert space). For $F, G \in \mathcal{A}_+$ define

$$\langle F, G \rangle_{\text{OS}} := \mathcal{S}(\Theta F G).$$

By reflection positivity this is positive semidefinite. Let $\mathcal{N} := \{F \in \mathcal{A}_+ : \langle F, F \rangle_{\text{OS}} = 0\}$ and set $\mathcal{H}_0 := \mathcal{A}_+ / \mathcal{N}$. The inner product descends to $\mathcal{H}_0$; let $\mathcal{H}$ be its completion. Denote by $[F] \in \mathcal{H}$ the class of $F \in \mathcal{A}_+$.

The vacuum vector is $\Omega := [1]$.

3.2 The OS semigroup

Lemma 3.2 (Pre-semigroup on the OS space). *Let $\mathcal{S}$ be OS-positive in the sense of Definition 2.3 and let $(\mathcal{H}, \Omega)$ be the associated OS Hilbert space. For $t \geq 0$, define on $\mathcal{H}_0$ the operator*

$$T_t[F] := [\tau_t F].$$

Then $(T_t)_{t \geq 0}$ is a well-defined semigroup of symmetric operators on $\mathcal{H}_0$.

Proof. Since $\tau_t \mathcal{A}_+ \subseteq \mathcal{A}_+$ for $t \geq 0$, the definition makes sense on $\mathcal{H}_0$.

Well-definedness. If $F \in \mathcal{N}$ then $\| [F] \|^2 = \mathcal{S}(\Theta(F)F) = 0$. Using translation invariance and the covariance $\Theta(\tau_t F) = \tau_{-t} \Theta(F)$,

$$\| [\tau_t F] \|^2 = \mathcal{S}(\Theta(\tau_t F) \tau_t F) = \mathcal{S}(\tau_{-t} \Theta(F) \tau_t F) = \mathcal{S}(\Theta(F) \tau_{2t} F).$$

By Cauchy–Schwarz for the OS inner product (valid for any positive semidefinite sesquilinear form),

$$|\mathcal{S}(\Theta(F) \tau_{2t} F)|^2 \leq \mathcal{S}(\Theta(F)F) \mathcal{S}(\Theta(\tau_{2t} F) \tau_{2t} F) = 0,$$

hence $\| [\tau_t F] \|^2 = 0$ and $\tau_t F \in \mathcal{N}$. Thus T_t is well-defined on the quotient.

Semigroup and symmetry. The semigroup property follows from $\tau_{t+s} = \tau_t \tau_s$. Moreover,

$$\langle T_t[F], [G] \rangle = \mathcal{S}(\Theta(\tau_t F) G) = \mathcal{S}(\tau_{-t} \Theta(F) G) = \mathcal{S}(\Theta(F) \tau_t G) = \langle [F], T_t[G] \rangle,$$

so T_t is symmetric on $\mathcal{H}_0$. □

Assumption 3.3 (OS reconstruction package: contraction semigroup and Hamiltonian). In the setting of Definition 2.3, we assume that the standard OS reconstruction applies to the time-translation operators defined on $\mathcal{H}_0$ in Lemma 3.2: the family $(T_t)_{t \geq 0}$ extends uniquely to a strongly continuous symmetric *contraction* semigroup on $\mathcal{H}$. Consequently there exists a self-adjoint operator $H \geq 0$ on $\mathcal{H}$ such that $T_t = e^{-tH}$ for all $t \geq 0$.

Strong continuity can be obtained, for example, from the weak continuity in (OS5) together with contractivity; see e.g. [7].

This reconstruction package is standard; for reflection-positive bounded-observable settings see, for example, Osterwalder–Schrader [5], Glimm–Jaffe [3], and Fröhlich–Israel–Lieb–Simon [6].

Lemma 3.4 (OS matrix element identity). *For $F, G \in \mathcal{A}_+$ and $t \geq 0$,*

$$\langle [F], e^{-tH} [G] \rangle = \mathcal{S}(\Theta F \tau_t G).$$

Proof. By Assumption 3.3, $e^{-tH} = T_t$ and $T_t[G] = [\tau_t G]$. Therefore

$$\langle [F], e^{-tH} [G] \rangle = \langle [F], [\tau_t G] \rangle_{\text{OS}} = \mathcal{S}(\Theta F \tau_t G).$$

□

4 Spectral-gap criteria for a positive semigroup

We state a usable spectral-theorem criterion in the quantifier form naturally produced by exponential clustering.

Lemma 4.1 (Spectral exclusion from exponential decay on a dense set). *Let $H \geq 0$ be self-adjoint on $\mathcal{H}$ and let P_0 denote the spectral projection onto $\{0\}$ (the ground-state sector). Let $\mathcal{D} \subset (\text{Ran } P_0)^\perp$ be a dense subspace. Suppose there exists $m > 0$ such that for every $\psi \in \mathcal{D}$ there is a constant $C_\psi < \infty$ with*

$$\langle \psi, e^{-tH} \psi \rangle \leq C_\psi e^{-mt} \quad \forall t \geq 0.$$

Then

$$\text{Spec}(H) \cap (0, m) = \emptyset.$$

Equivalently, $\text{Spec}(H) \subset \{0\} \cup [m, \infty)$.

Proof. Fix $0 < \lambda < m$ and let $P_{[0, \lambda]}$ denote the spectral projector of H onto $[0, \lambda]$. It suffices to show $P_{[0, \lambda]} = 0$.

Let $\psi \in \mathcal{D}$. Decompose $\psi = \psi_{\leq \lambda} + \psi_{> \lambda}$ where $\psi_{\leq \lambda} = P_{[0, \lambda]} \psi$. Since $\psi \perp \text{Ran } P_0$, we have $P_{\{0\}} \psi = 0$, hence $\psi_{\leq \lambda} = P_{(0, \lambda]} \psi$.

By the spectral theorem,

$$\langle \psi, e^{-tH} \psi \rangle \geq \langle \psi_{\leq \lambda}, e^{-tH} \psi_{\leq \lambda} \rangle \geq e^{-\lambda t} \|\psi_{\leq \lambda}\|^2.$$

Combining with the assumed upper bound gives

$$\|\psi_{\leq \lambda}\|^2 \leq C_\psi e^{-(m-\lambda)t} \quad \forall t \geq 0.$$

Letting $t \rightarrow \infty$ yields $\psi_{\leq \lambda} = 0$. Thus $P_{(0, \lambda]} \psi = 0$ for all $\psi \in \mathcal{D}$. By density of $\mathcal{D}$ and boundedness of $P_{(0, \lambda]}$, it follows that $P_{(0, \lambda]} = 0$. Since $\lambda < m$ was arbitrary, $\text{Spec}(H) \cap (0, m) = \emptyset$. $\square$

5 From Euclidean exponential clustering to a spectral gap

We now connect Euclidean clustering bounds (a natural output of Dobrushin-type contractivity) to the hypothesis of Lemma 4.1.

5.1 Exponential decay hypothesis

Definition 5.1 (Euclidean exponential time clustering on $\mathcal{A}_+$). Let $\mathcal{S}$ be an OS-positive translation-invariant functional. We say $\mathcal{S}$ has *exponential time clustering with rate $m > 0$* if for every $F, G \in \mathcal{A}_+$ there exists $C_{F,G} < \infty$ such that

$$|\mathcal{S}(\Theta F \tau_t G) - \mathcal{S}(\Theta F) \mathcal{S}(G)| \leq C_{F,G} e^{-mt} \quad \forall t \geq 0.$$

Remark 5.2. In lattice Gibbsian settings, exponential clustering is often first established for truncated correlations of bounded local observables separated in Euclidean distance, for example under Dobrushin uniqueness/contractivity; see [1, 2]. Definition 5.1 isolates the time-separated specialization needed for semigroup spectral extraction. The constants are allowed to depend on the observables. A standard geometric reduction from distance clustering to time clustering is recalled in Remark 5.4.

Lemma 5.3 (Centered vectors are dense in $\Omega^\perp$). *Let $(\mathcal{H}, \Omega)$ be the OS Hilbert space associated to $\mathcal{S}$. Then*

$$\overline{\text{span}\{[\tilde{F}] : F \in \mathcal{A}_+\}} = \Omega^\perp, \quad \tilde{F} := F - \mathcal{S}(F)\mathbf{1}.$$

Proof. By construction, $\text{span}\{[F] : F \in \mathcal{A}_+\}$ is dense in $\mathcal{H}$. For $F \in \mathcal{A}_+$ we have

$$[F] = \langle \Omega, [F] \rangle \Omega + ([F] - \langle \Omega, [F] \rangle \Omega) = \mathcal{S}(F)\Omega + [F - \mathcal{S}(F)\mathbf{1}].$$

The second summand lies in $\Omega^\perp$ and equals $[\tilde{F}]$. Taking closures yields the claim. $\square$

Remark 5.4 (From distance clustering to time clustering). In applications, exponential time clustering is typically deduced from exponential decay of truncated correlations in Euclidean distance: for bounded local observables A, B localized in regions Λ_A, Λ_B one has

$$|\mathcal{S}(AB) - \mathcal{S}(A)\mathcal{S}(B)| \leq C_{A,B} e^{-m \text{dist}(\Lambda_A, \Lambda_B)}.$$

Applying this with $A = \Theta F$ and $B = \tau_t G$ yields Definition 5.1 because $\text{dist}(\theta\Lambda_F, \Lambda_G + (t, \mathbf{0}))$ grows linearly in t for fixed bounded $\Lambda_F, \Lambda_G \subset \mathbb{R}_+^4$. See, for example, [1, 2] for general treatments of exponential clustering for Gibbs fields under Dobrushin-type assumptions.

5.2 Entrance-to-gap theorem (spectral extraction)

Theorem 5.5 (Spectral extraction on a bounded macroscopic net). *Let $\mathcal{S}$ be an OS-positive translation-invariant functional on a bounded macroscopic net, with reconstructed $(\mathcal{H}, \Omega, H)$. Assume that $\mathcal{S}$ satisfies exponential time clustering with rate $m > 0$ in the sense of Definition 5.1. Then*

$$\text{Spec}(H) \subset \{0\} \cup [m, \infty).$$

Proof. Let $\mathcal{D} := \text{span}\{[\tilde{F}] : F \in \mathcal{A}_+\}$ where $\tilde{F} := F - \mathcal{S}(F)\mathbf{1}$ is the centered observable. Then $[\tilde{F}] \perp \Omega$ because

$$\langle \Omega, [\tilde{F}] \rangle = \mathcal{S}(\Theta \mathbf{1} \tilde{F}) = \mathcal{S}(\tilde{F}) = 0.$$

By Lemma 5.3, $\mathcal{D}$ is dense in $\Omega^\perp$.

Step 1: decay on generators. Fix $\psi = [\tilde{F}] \in \mathcal{D}$. By Lemma 3.4,

$$\langle \psi, e^{-tH} \psi \rangle = \mathcal{S}(\Theta \tilde{F} \tau_t \tilde{F}).$$

By exponential time clustering,

$$|\mathcal{S}(\Theta \tilde{F} \tau_t \tilde{F})| = |\mathcal{S}(\Theta F \tau_t F) - \mathcal{S}(\Theta F) \mathcal{S}(F)| \leq C_{F,F} e^{-mt}.$$

Since the left side is nonnegative (it is a semigroup quadratic form), we obtain

$$\langle \psi, e^{-tH} \psi \rangle \leq C_{F,F} e^{-mt} \quad \forall t \geq 0.$$

Step 2: decay on finite linear combinations. Let $\psi = \sum_{i=1}^n c_i [\tilde{F}_i] \in \mathcal{D}$. Then

$$\langle \psi, e^{-tH} \psi \rangle = \sum_{i,j=1}^n \bar{c}_i c_j \mathcal{S}(\Theta \tilde{F}_i \tau_t \tilde{F}_j).$$

For each pair (i, j) , Definition 5.1 applied to $\tilde{F}_i, \tilde{F}_j \in \mathcal{A}_+$ yields a constant $C_{ij} < \infty$ such that $|\mathcal{S}(\Theta \tilde{F}_i \tau_t \tilde{F}_j)| \leq C_{ij} e^{-mt}$ for all $t \geq 0$. Therefore, using the triangle inequality,

$$0 \leq \langle \psi, e^{-tH} \psi \rangle \leq \sum_{i,j=1}^n |c_i| |c_j| C_{ij} e^{-mt} =: C_\psi e^{-mt}.$$

Step 3: apply the spectral exclusion lemma. Let P_0 denote the spectral projection onto $\{0\}$. Since $H \geq 0$ is self-adjoint, the spectral theorem gives $e^{-tH} \rightarrow P_0$ strongly as $t \rightarrow \infty$. The decay bound from Step 2 therefore implies $\|P_0\psi\|^2 = \lim_{t \rightarrow \infty} \langle \psi, e^{-tH}\psi \rangle = 0$ for all $\psi \in \mathcal{D}$. Hence $\mathcal{D} \subset (\text{Ran } P_0)^\perp$. Since $\Omega \in \text{Ran } P_0$ we have $(\text{Ran } P_0)^\perp \subset \Omega^\perp$, while the inclusion $\Omega^\perp \subset (\text{Ran } P_0)^\perp$ follows from $\mathcal{D} \subset (\text{Ran } P_0)^\perp$ and density of $\mathcal{D}$ in $\Omega^\perp$. Hence $(\text{Ran } P_0)^\perp = \Omega^\perp$, and in particular $\mathcal{D}$ is dense in $(\text{Ran } P_0)^\perp$. Thus the hypotheses of Lemma 4.1 hold, yielding $\text{Spec}(H) \cap (0, m) = \emptyset$. $\square$

Corollary 5.6 (Uniqueness of the vacuum in the reconstructed representation). *Under the hypotheses of Theorem 5.5, the ground-state sector is one-dimensional: $\ker H = \mathbb{C}\Omega$.*

Proof. Let P_0 be the spectral projection onto $\{0\}$. For any $\psi \in \mathcal{D}$ the exponential bound in the proof of Theorem 5.5 implies $\langle \psi, e^{-tH}\psi \rangle \rightarrow 0$ as $t \rightarrow \infty$. By the spectral theorem, $e^{-tH} \rightarrow P_0$ strongly, hence $\langle \psi, e^{-tH}\psi \rangle \rightarrow \|P_0\psi\|^2$. Therefore $P_0\psi = 0$ for all $\psi \in \mathcal{D}$. Since $\mathcal{D}$ is dense in $\Omega^\perp$, this implies $P_0 \upharpoonright \Omega^\perp = 0$, and hence $\text{Ran } P_0 \subset (\Omega^\perp)^\perp = \mathbb{C}\Omega$. Because $\Omega \in \text{Ran } P_0$, we conclude $\text{Ran } P_0 = \mathbb{C}\Omega$, i.e. $\ker H = \mathbb{C}\Omega$. $\square$

6 Discussion: “time” in spectral extraction

Theorem 5.5 is the abstract interface used in macroscopic mass-gap constructions:

1. a Euclidean argument yields OS positivity on a bounded local net;
2. a contraction or mixing argument yields exponential time clustering;
3. OS reconstruction converts time translations into a semigroup e^{-tH} ;
4. exponential decay of Euclidean matrix elements forces $\text{Spec}(H) \subset \{0\} \cup [m, \infty)$.

No renormalization-group flow parameter is required for the spectral conclusion. Any RG procedure (stable-manifold or finite-depth entrance) is, from this point of view, a means to obtain exponential clustering at some macroscopic scale. The spectral extraction itself is semigroup-theoretic.

A A short spectral-theorem lemma

For completeness we record a standard variant used implicitly above.

Lemma A.1 (Lower spectral weight forces slow decay). *Let $H \geq 0$ be self-adjoint and let $\psi \in \mathcal{H}$ with spectral measure μ_ψ . If $\mu_\psi((0, \lambda]) > 0$ for some $\lambda > 0$, then*

$$\langle \psi, e^{-tH}\psi \rangle \geq e^{-\lambda t} \mu_\psi((0, \lambda]) \quad \forall t \geq 0.$$

Proof. By the spectral theorem, $\langle \psi, e^{-tH}\psi \rangle = \int_{[0, \infty)} e^{-ts} \mu_\psi(ds)$. Restricting the integral to $(0, \lambda]$ gives the bound. $\square$

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